CMPE491 – Senior Project I

Project Specifications Report

HorusEye – AI-Based Exam Proctoring System

7 November 2025

Çağla Abazaoğlu – 10051217210

Gizem Nur İpek – 46162949900

Taha Kürşat Öztürk –15709032030

Ali Sahil – 14596166750

Tuğba Hilal Kırer – 10015328662

Contents

1. Introduction

Determining and implementing a great graduation project is a very critical issue for the shaping of future careers of final-year students in the Computer Engineering major. Our team, made up of people who have the same ethical values, principles, and long-term vision, decided to develop HorusEye, an AI-based exam proctoring system to ensure fairness using computer vision and machine learning in both online and on-campus examinations.

Increasing adoption of hybrid and remote education models is driving the demand for reliable, automated, and scalable proctoring solutions. Traditional human-based invigilation usually suffers from a plethora of drawbacks, such as the occurrence of human error, subjective judgments, and high operational costs. In this regard, the HorusEye system aims to bridge this gap by offering an intelligent, real-time monitoring solution capable of detecting suspicious behaviors during exams and alerting supervisors in an instant.

This document describes the details of the system, including the description of the system, constraints, ethical considerations, and requirements.

1.1 Description

HorusEye is designed as a computer vision–driven surveillance system that detects and reports probable cheating behavior during examinations. The system does so by using one or more cameras strategically placed in an exam room to continuously analyze student activity with the application of AI algorithms.

The features include:

- Real-time monitoring with deep-learning-based behavioral detection models.
- Detection of suspicious movements, abnormal eye shifts, whispering, use of mobile devices, or sharing of materials.
- Automatic alerts sent to exam supervisors through a secure web dashboard.
- Incident logging with time-stamped video evidence for later review.
- Post-exam analysis reports that summarize flagged behaviors and patterns.

The front-end will be developed in either React.js or Vue.js, depending on the final UI component and state management requirements. System communication will be controlled via RESTful API endpoints, while real-time alerts and exam event streaming will be facilitated over WebSocket channels. The back-end architecture will be designed as Python-based microservices, wherein the tasks related to computer vision-face detection, gaze tracking, and object recognition-will be accomplished with OpenCV (with optional C++ acceleration) and YOLOv8. Anomaly detection over time will be performed with TensorFlow-based LSTM/GRU models. Deployment and scaling will be done with Docker containers, and Redis will be used for real-time caching and state synchronization. All persistent data shall be securely stored in a PostgreSQL database, optionally managed via Supabase for cloud-based orchestration.

This will ensure academic integrity while offering universities a scalable digital invigilation solution that is adaptable to both physical and online exams.

1.2 Constraints

During the stages of planning and design, there were several constraints noted:

- Hardware Limitations: Camera quality and frame rate directly affect the precision of AI detection.
- Time Limitations: Development and testing should be done within one academic semester.
- Privacy & Data Protection: Any capturing of video data needs to be GDPR/KVKK-compliant for the protection of student privacy.
- Budget Limitation: Cloud services and GPU usage for AI training are limited to free or student-licensed plans.
- Environmental Variability: Lighting and seating arrangements in different exam halls would affect performance and accuracy.

1.3 Professional and Ethical Issues

AI-based monitoring will therefore need to be developed with due consideration for the ethical issues involved. Recording and analyzing student behavior raises significant concerns about privacy and consent. All data collection must be carried out in accordance with institutional policies and national data protection laws.

Additionally, bias in AI models, such as misinterpreting natural gestures, should be reduced to a minimum by using diverse training datasets. The algorithm design shall place a strong focus on transparency, explainability, and fairness so that no student is treated unjustly.

2. Requirements

Functional Requirements:

- The system shall detect suspicious behaviors: eye movements, phone usage, communication.
- The system shall generate instantaneous alerts and notifications for proctors.
- The system shall record incidents with timestamps and store them along with relevant video clips.
- The system shall generate summarized post-exam reports.
- The system shall operate in both modes: real-time monitoring and offline analysis.

Non-Functional Requirements:

- The application shall operate on Windows, macOS, and Linux environments.
- The system shall ensure secure data transmission, using HTTPS.
- The system shall achieve at least 85% detection accuracy under standard lighting conditions.
- The system should comply with the privacy and ethical standards defined by the university.
- The user interface shall be intuitive and responsive for proctors and administrators.